

Convergent Concept Representations

A Learned, From-Scratch Vector-Symbolic Encoder with Wave-Domain Binding and Calibrated Confidence: Theory, Capacity Bounds, and a Falsifiable Research Program

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Abstract

Two surface forms of the same concept — a paraphrase, an augmentation, ultimately a different modality — should map to nearly the same internal representation, and a model should be able to *state how sure it is* and *show its work*. We present the theory and a falsifiable research program for a learned, from-scratch encoder (working name **Convergent Concept Representations**, CCR) that maps raw inputs to sparse distributed **concept patterns** stored as **attractors** — so the same concept re-triggers nearly the same pattern — and predicts the *next pattern* in that shared space rather than the next discrete vocabulary symbol. The representational substrate is that of **Vector Symbolic Architectures / Hyperdimensional Computing** (Kanerva; Plate; Smolensky [25,26,27,29]): high-dimensional, near-orthogonal codes combined by three algebraic operations — **bind**, **superpose**, **permute**. Our central mechanistic claim is that the "wave" (complex-amplitude) mathematics is *not a separate device* but the frequency-domain form of vector-symbolic binding: circular-convolution binding, the canonical VSA bind, is elementwise complex multiplication — a magnitude product and a **phase addition** — so a complex-amplitude representation read out by the **Born rule** is a learnable, end-to-end-differentiable realization of VSA binding. We contribute: (i) this binding identity, plus a proof that unitary evolution on $\mathbb{C}^d$ is structured orthogonal evolution on $\mathbb{R}^{2d}$ and hence trainable by ordinary autodiff; (ii) an **invariance objective** (predict the next pattern under a stop-gradient target), its representation-collapse failure mode, and three anti-collapse regularizers; (iii) a capacity analysis separating *naming* capacity (combinatorial and Johnson–Lindenstrauss bounds, exponential in dimension) from *storage* capacity (classical Hopfield **0.138N** vs. modern Hopfield's exponential capacity), resolving an order-of-magnitude fallacy; (iv) a Bayesian–Hebbian salience ("glow") layer yielding a **calibrated, inspectable confidence** signal, with a mandatory stabilization against frequency runaway; and (v) a complete parameter, memory, and FLOP budget showing every *decision-grade* experiment fits an Apple M1 (a **12.6M**-parameter model, $\sim$ **151 MB** training footprint, **2.3** $\times$ smaller than a parameter-matched token baseline). The **testable** headline is *single-modality invariance* — the same concept under paraphrase / augmentation / byte-noise maps to nearly the same pattern — with cross-modal binding ("one pattern for an image, a sentence, or a video") stated as a scaling destination, **not** a demonstrated result. We argue separately, and quantitatively, that a fault-tolerant quantum computer is the *wrong* tool for this workload (barren plateaus $\text{Var}[\partial_\theta L] \sim 2^{-\alpha n}$, dequantization, the simulability pincer), which is precisely why every component runs

classically. The empirical precedent — a single complex-valued layer reaching **90.91%/91.66%** on AG News versus $\sim 71.7\%$ for a single Transformer layer [1] — is treated as motivation to be re-established under strict fairness controls, not a result we inherit.

Keywords: vector symbolic architectures, hyperdimensional computing, holographic reduced representations, complex-valued neural networks, modality-invariant representation, associative memory, modern Hopfield networks, representation collapse, calibration.

1. Introduction

1.1 The token paradigm and its hidden costs

Contemporary large language models factorize text autoregressively over a fixed vocabulary V , maximizing $\sum_n \log p(x_{n+1} | x_{\leq n})$ where each x is a discrete symbol. Two structural costs of this paradigm are rarely stated as costs:

1. **A one-hot learning signal.** The training target at each position carries at most $\log_2 |V|$ bits and, as a gradient signal, is a single "correct class" among $|V|$. The model's continuous internal state is supervised only through this discrete bottleneck.
2. **A vocabulary tax on parameters.** The input/output embedding tables occupy $|V| \cdot d$ parameters. At small scale this is not a rounding error: for $|V| = 32000$, $d = 512$ it is 16.4M parameters, which we show below is 39–57% of a 4–8 layer model (§4.4).

Neither cost is fundamental to *language*; both are artifacts of choosing the discrete symbol as the unit of prediction.

1.2 The problem, and the base we test

The problem we target is a **stable, surface-form-invariant, inspectable concept representation**: the same concept — under paraphrase, augmentation, noise, and eventually a different modality — should evoke nearly the same **distributed pattern**, that pattern should be a reliable **attractor** (re-triggering the same nodes), and the model should expose a **calibrated confidence** in its retrieval. This is the representational problem underneath grounding, retrieval, cross-modal alignment, and on-device anti-hallucination; it is not specific to language.

The base we test is that of **Vector Symbolic Architectures (VSA) / Hyperdimensional Computing (HDC)** (Kanerva [27]; Plate [26]; Smolensky [29]; survey [25]): concepts are high-dimensional, near-orthogonal vectors, and structure is built from three algebraic operations — **bind** (associate a role with a filler), **superpose** (form a set), and **permute** (encode order/position). This is *exactly* the intuition that "a fixed but expandable substrate whose nodes and weights combine combinatorially produces stable, reusable patterns" — permutation is a named VSA primitive, not a metaphor. VSA gives us capacity theorems (§5) and a rigorous home; what we add is that the substrate is **learned end-to-end and from scratch** (§C1), stored as

attractors with modern-Hopfield retrieval (§5.4), predicted with an **invariance objective** (§4), and equipped with a **calibrated confidence** layer (§6).

The role of the **complex-amplitude ("wave") mathematics** is then precise and demoted from identity to mechanism: it is the **frequency-domain implementation of VSA binding** (§3.3). The one representational novelty over a real-valued net is **phase** — the orthogonal degree of freedom that supplies *interference* (two equal-magnitude patterns reinforce or cancel by their relative phase) — and whether phase earns its keep at equal parameters is a single pre-registered ablation (H1, §9), not an assumption. "Quantum" here names only the **mathematics** (Hilbert-space states, unitary maps, Born-rule readout) executed on classical hardware; §7 argues that running classically is the correct engineering choice on the merits, not a compromise.

1.3 Contributions and scope

This is a **theory-and-protocol** paper. Its contributions are the derivations of §3, the objective analysis of §4, the capacity theorems of §5, the calibration mechanism of §6, the quantitative case against quantum hardware in §7, the full resource accounting of §8, and the falsification design of §9. Its scope is deliberately bounded: we make claims at **small scale** (classification and character/byte-level modeling on an M1) and treat **generation at scale** as unproven, gated behind an explicit readout-cost wall (§8.2). We do not claim, and our design cannot support, a head-to-head against frontier models; §9 defines the only comparisons we consider valid.

All experimental commitments obey a purity charter (no pretrained components, raw-byte defaults, built-from-scratch parameter-matched baselines, bit-for-bit determinism) detailed in the companion `docs/GROUND-UP-CONSTRAINTS.md` ; here we state results as *hypotheses with kill numbers*.

2. Background and related work

Complex-valued and quantum-inspired language models. The *Wave Network* [1] represents each token as a complex vector whose magnitude encodes global semantics and whose phase encodes the token–context relationship, updated by interference or modulation; a *single* such layer attains **90.91%** (interference) and **91.66%** (modulation) on AG News, exceeding a single Transformer layer built on BERT embeddings by **19.23** and **19.98** points and approaching fine-tuned BERT-base (**94.64%**), using a $\sim$ **2.4M**-parameter model and reducing memory/time by **77.34%/85.62%** versus BERT-base. Its backward dynamics are worked out in [2]. *Complex-valued Neural Network-based Quantum Language Models* [3] model semantic units (sememe $\rightarrow$ word $\rightarrow$ sentence) on a single Semantic Hilbert Space with complex-valued composition and report complex outperforming real-valued counterparts across six classification datasets. These are our closest precedents; we re-establish the *comparison* under fairness controls (§9) because [1]’s reference used pretrained embeddings.

Unitary and tensor-network models. Unitary-evolution RNNs [4] constrain the recurrent operator to be unitary, preserving hidden-state norm and solving copy/long-memory tasks that LSTMs fail. Matrix-product-

state classifiers [5] give exponential parameter compression for low-bond-dimension weight tensors. Both are classical demonstrations that quantum *formalism* buys concrete efficiency.

Prediction in latent/continuous space. *Coconut* [6] feeds a model's last hidden state back as the next input embedding, reasoning in a continuous latent space that can encode multiple next steps and perform an emergent breadth-first search, improving token efficiency on planning-heavy reasoning. *Large Concept Models* [7] predict in a sentence-embedding space (SONAR) and outperform same-size LLMs zero-shot across many languages. LeCun's **JEPA** program predicts *representations* under a stop-gradient/EMA target rather than reconstructing inputs. These motivate pattern-as-token; where they use pretrained embedding spaces (e.g. SONAR) we quarantine those and learn the space from scratch.

Vector symbolic architectures and distributed binding. VSA/HDC [25] represents symbols as high-dimensional, near-orthogonal vectors combined by three operations — bind, superpose, permute. The family includes Smolensky's tensor-product representations [29], Plate's **Holographic Reduced Representations** [26] (binding by circular convolution), Gayler's Multiply-Add-Permute, and Kanerva's Sparse Distributed Memory and Binary Spatter Codes [27]. This is the field our substrate lives in, and the frequency-domain view of HRR binding is exactly our complex-amplitude "wave" mechanism (§3.3). The distinction from classical VSA is that its codebooks and binding maps are largely *hand-designed and un-calibrated*; our object is a **learned-from-scratch** VSA-style encoder with attractor storage and a calibrated confidence layer. This is the closest prior-art field to the present work and, to our knowledge, is under-connected to the complex-valued / "wave" language-model line [1,3]; §3.3 makes the connection explicit.

Modality-invariant and shared-embedding spaces. CLIP-style contrastive alignment and **ImageBind** [28] demonstrate that many modalities can be bound into a single embedding space, and Large Concept Models [7] that predicting in such a space can beat same-size token LLMs. These rely on strong *pretrained* per-modality encoders; under our no-pretrained charter (C1) the cross-modal version is a stated scaling destination (§10), and the claim we actually test is single-modality invariance.

Associative memory. Classical Hopfield networks [8] store $\approx 0.138N$ patterns in N neurons [9]. *Modern* Hopfield networks [10] use a log-sum-exp energy with **exponential** storage capacity in the representation dimension, retrieve in one update, and have an update rule identical to Transformer attention — the bridge that makes "concepts as attractors" computationally cheap (§5).

Why not quantum hardware. Barren plateaus [11] show the gradient variance of wide parameterized quantum circuits vanishes exponentially in qubit number. Dequantization [12] shows many claimed QML speedups have classical algorithms only polynomially slower. The one rigorously proven advantage [13] is for a discrete-log-structured dataset in the data-scarce regime. The simulability pincer [14] collects evidence that circuits provably free of barren plateaus are also classically simulable. §7 develops these quantitatively.

3. The representation: distributed patterns and wave-domain binding

We write a state as $\psi \in \mathbb{C}^d$, stored in implementation as a pair of real tensors $(\mathbf{Re} \psi, \mathbf{Im} \psi) \in \mathbb{R}^d \times \mathbb{R}^d$. The Hermitian inner product is $\langle a | b \rangle = \sum_k \overline{a_k} b_k$, and $\|\psi\|^2 = \langle \psi | \psi \rangle$. A *concept pattern* is such a vector; the three VSA operations (§1.2) are superposition (§3.2, addition), binding (§3.3, below), and permutation (a fixed invertible reindexing P , $\psi \mapsto P\psi$, used to protect order — omitted from the equations for brevity as it is a special orthogonal map covered by §3.4).

3.1 Amplitudes and the Born rule

A single amplitude is $\alpha = r e^{i\phi}$ with magnitude $r \geq 0$ and phase $\phi \in [0, 2\pi)$. Given a normalized state ($\|\psi\| = 1$) and an orthonormal measurement basis $\{|e_k\rangle\}$, the **Born rule** assigns outcome probabilities

$$p_k = |\langle e_k | \psi \rangle|^2, \quad \sum_k p_k = \sum_k |\langle e_k | \psi \rangle|^2 = \|\psi\|^2 = 1.$$

Normalization is therefore not a convenience but the statement that the state lives on the unit sphere $S^{2d-1} \subset \mathbb{C}^d$, which pairs naturally with norm-preserving (unitary) dynamics (§3.4).

3.2 Interference (the algebra that phase buys)

Consider superposing two amplitudes $\psi_j = r_j e^{i\phi_j}$ (scalar case; the vector case is coordinatewise). The Born-rule intensity of the sum is derived by expanding the modulus-squared:

$$|\psi_1 + \psi_2|^2 = (\psi_1 + \psi_2) \overline{(\psi_1 + \psi_2)} = |\psi_1|^2 + |\psi_2|^2 + \psi_1 \overline{\psi_2} + \overline{\psi_1} \psi_2.$$

Since $\psi_1 \overline{\psi_2} = r_1 r_2 e^{i(\phi_1 - \phi_2)}$ and $z + \bar{z} = 2 \operatorname{Re}(z)$,

$$\boxed{|\psi_1 + \psi_2|^2 = r_1^2 + r_2^2 + \underbrace{2 r_1 r_2 \cos(\phi_1 - \phi_2)}_{\text{interference term}}}$$

The interference term depends **only on the relative phase** $\Delta\phi = \phi_1 - \phi_2$:

$\Delta\phi$	$\cos \Delta\phi$	intensity	interpretation
0	+1	$(r_1 + r_2)^2$	constructive (agreement/reinforcement)
$\pi/2$	0	$r_1^2 + r_2^2$	independent (classical addition)
π	-1	$(r_1 - r_2)^2$	destructive (e.g. negation/cancellation)

Equal-magnitude units ($r_1 = r_2 = r$) either quadruple in intensity ($4r^2$) or annihilate (0) purely as a function of $\Delta\phi$. A real-valued network must *learn* cancellation with additional nonlinear capacity; a complex model obtains it in the algebra. Whether this pays off at equal parameters is hypothesis **H1** (§9).

3.3 Modulation (multiplicative binding)

The Hadamard (element-wise) product binds two states multiplicatively:

$$\psi_1 \odot \psi_2 = (r_1 \odot r_2) e^{i(\phi_1 + \phi_2)},$$

multiplying magnitudes and **adding** phases. Interference is additive superposition (phases can cancel); modulation is a gating/binding operation (phases accumulate). Both are $O(d)$ and both were competitive in [1].

Proposition 0 (wave modulation is frequency-domain VSA binding). The canonical VSA *bind* in Holographic Reduced Representations [26] is circular convolution $a \otimes b$. By the circular convolution theorem, the discrete Fourier transform diagonalizes it:

$$\mathcal{F}(a \otimes b) = \mathcal{F}(a) \odot \mathcal{F}(b),$$

i.e. binding is **elementwise complex multiplication** of the spectra. Writing each spectral coordinate in polar form $\mathcal{F}(a)_k = \rho_k e^{i\theta_k}$, this multiplies magnitudes and **adds phases** — exactly the modulation of §3.3. Two consequences follow. (i) Working *directly* in the amplitude ("wave") domain, our modulation **is** VSA binding; a phase-only (unit-magnitude) state is Plate's *circular* HRR, and the Born-normalized state of §3.1 is its unit-norm case. (ii) Unbinding (querying a bound pair) is the inverse operation, circular correlation, which in the spectral domain is multiplication by the complex **conjugate** — a phase *subtraction*. So bind/unbind, superpose, and permute — the full VSA algebra — are all available as $O(d)$ (or $O(d \log d)$ via FFT) complex operations, and, unlike classical hand-designed VSA, the codebook and the maps here are **learned** (§4) and read out by the Born rule (§3.4–§4). This identity is why the complex mathematics is a *mechanism* for a distributed-binding representation, not a separate thesis: dropping it (the `phase=0` ablation, H1) reduces the model to a real-valued distributed code and asks whether phase-domain binding earned its parameters.

3.4 Unitary evolution and its real form

We evolve states by a linear map $\psi' = U\psi$ constrained to be **unitary**, $U^\dagger U = I$, so that $\|\psi'\| = \|\psi\|$ (Born normalization is preserved and, as in [4], gradients neither vanish nor explode across depth/time).

Proposition 1 (unitary $\mathbb{C}^d =$ structured orthogonal $\mathbb{R}^{2d}$). Write $U = A + iB$ with $A, B \in \mathbb{R}^{d \times d}$, and represent $\psi = x + iy$ by the stacked real vector $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^{2d}$. Then complex multiplication by U is the real linear map

$$M = \begin{pmatrix} A & -B \\ B & A \end{pmatrix}, \quad \text{and} \quad U^\dagger U = I \iff M^\top M = I_{2d}.$$

Proof. $U^\dagger U = (A^\top - iB^\top)(A + iB) = (A^\top A + B^\top B) + i(A^\top B - B^\top A)$, so $U^\dagger U = I$ iff $A^\top A + B^\top B = I$ and $A^\top B = B^\top A$. Compute

$$M^\top M = \begin{pmatrix} A^\top & B^\top \\ -B^\top & A^\top \end{pmatrix} \begin{pmatrix} A & -B \\ B & A \end{pmatrix} = \begin{pmatrix} A^\top A + B^\top B & -A^\top B + B^\top A \\ -B^\top A + A^\top B & B^\top B + A^\top A \end{pmatrix}.$$

This equals I_{2d} under exactly the same two conditions. ■

Consequently a unitary complex layer is a real orthogonal layer with tied 2×2 block structure — implementable with real tensors and differentiated by ordinary backpropagation. We parameterize U three ways:

- **Matrix exponential:** $U = e^{iH}$, $H = H^\dagger$. Exact but $O(d^3)$ (eigendecomposition) and awkward to differentiate stably.
- **Cayley transform:** $U = (I - A)(I + A)^{-1}$ with A skew-Hermitian ($A^\dagger = -A$). One $O(d^3)$ solve; numerically friendly; preferred on CPU.
- **Givens/Householder products:** $U = \prod_m G_m$ of 2-D rotations/reflections; $O(d)$ each, **inverse-free**, preferred on Metal/GPU.

Proposition 2 (Cayley is unitary). For skew-Hermitian A , $U = (I - A)(I + A)^{-1}$ satisfies $U^\dagger U = I$. *Proof.* $(I + A)^\dagger = I - A$ and $(I - A)^\dagger = I + A$, so $U^\dagger = (I - A)^{-1}(I + A)$. Because $(I - A)$ and $(I + A)$ are polynomials in A they commute, hence $U^\dagger U = (I - A)^{-1}(I + A)(I - A)(I + A)^{-1} = (I - A)^{-1}(I - A)(I + A)(I + A)^{-1} = I$. ■ (The transform is well defined whenever $-1 \notin \text{spec}(A)$; we bound the spectrum of A to keep $(I + A)$ well-conditioned.)

3.5 Training: Wirtinger gradients, no quantum taxes

For a real loss L of complex variables, the relevant update direction is the **Wirtinger derivative** $\partial L / \partial \bar{z}$, and $z \leftarrow z - \eta \partial L / \partial \bar{z}$. This is native in autodiff frameworks with complex support and, on the **(Re, Im)** representation, is *literally* real-valued autodiff. We therefore pay **none** of the quantum-hardware taxes: no parameter-shift gradient estimation, no measurement shot noise, and — critically (§7.1) — no barren plateau. The only overhead is a constant factor from complex arithmetic (§8.1).

4. Pattern-as-token: the predictive objective

4.1 Two objectives, contrasted

Next-token minimizes cross-entropy against a one-hot target:

$$L_{\text{token}} = -\log p(x_{n+1} \mid x_{\leq n}), \quad p(\cdot \mid x_{\leq n}) \in \Delta^{|V|-1}.$$

Next-pattern predicts the continuous latent of the next unit and matches it to a stop-gradient target in the *same* space:

$$L_{\text{pattern}} = D(\hat{z}_{n+1}, \text{sg}(z_{n+1})),$$

where $z_{n+1} = f_{\theta}(\text{unit}_{n+1})$ is the encoder's pattern, $\hat{z}_{n+1} = g_{\theta}(x_{\leq n})$ the predictor's guess, D a distance/similarity, and $\text{sg}(\cdot)$ the stop-gradient. The target lives in pattern space, so prediction is dense: every coordinate of z carries gradient, rather than the single "which symbol" bit of L_{token} .

4.2 The collapse failure mode (formal)

L_{pattern} admits a **trivial minimizer**: any constant encoder $f_{\theta} \equiv c$ makes $\hat{z} = z = c$ and $D = 0$, learning nothing. More generally the objective is minimized by any **low-rank** degenerate solution. We therefore monitor the **effective rank** (participation ratio) of the matrix $Z \in \mathbb{R}^{n \times d}$ of predicted patterns over a held-out set. If $\sigma_1 \geq \dots \geq \sigma_d \geq 0$ are its singular values,

$$\text{erank}(Z) = \frac{(\sum_i \sigma_i)^2}{\sum_i \sigma_i^2} \in [1, d],$$

with **erank** = 1 signalling full collapse and **erank** = d an isotropic code. A loss curve plunging toward 0 is, absent a rank guarantee, **evidence of collapse, not success** — this is the single most common way the objective lies to the experimenter, and instrumentation for it is wired *before* any training run (§9, H3).

4.3 Three anti-collapse regularizers

We treat all three as switchable and comparable:

(a) JEPA + VICReg. An **EMA target encoder** $\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau)\theta$ produces the (stop-gradient) target, preventing target/predictor co-collapse. VICReg adds explicit variance and covariance terms:

$$L = \underbrace{D(\hat{z}, \text{sg } z)}_{\text{invariance}} + \lambda_v \sum_{j=1}^d \max(0, \gamma - \sqrt{\text{Var}(z_{\cdot j}) + \epsilon}) + \lambda_c \sum_{i \neq j} [\text{Cov}(Z)]_{ij}^2.$$

The variance hinge forces each dimension's std above γ (defeating the constant solution); the covariance term decorrelates dimensions (defeating low-rank).

(b) InfoNCE (contrastive). With temperature τ and negatives $\{z^-\}$,

$$L_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(\hat{z}, z^+)/\tau)}{\sum_{z^-} \exp(\text{sim}(\hat{z}, z^-)/\tau)},$$

whose uniformity term makes constant outputs high-loss by construction.

(c) VQ codebook. Quantize targets to a learned codebook of K codes (EMA updates + dead-code reinitialization); the predictor targets a code index, which cannot collapse without collapsing codebook usage (monitored via codebook perplexity). A side benefit is cheap readout over $K \ll |V|$ codes (§8.2).

Selection rule (H2/H3): whichever regularizer yields no collapse at the best perplexity at equal parameters proceeds; if *none* avoids collapse within the perplexity budget, the project stops.

4.4 Parameter reclamation: a general calculation

Consider a decoder-style body of L layers at model width d . Standard per-layer parameter count is

$$\underbrace{4d^2}_{W_Q, W_K, W_V, W_O} + \underbrace{8d^2}_{\text{FFN } (d \rightarrow 4d \rightarrow d)} = 12d^2,$$

so the body has $12Ld^2$ parameters (layer-norm and biases are $O(d)$, negligible). A token model additionally carries a vocabulary table $|V|d$ (tied input/output). Hence the fraction of parameters spent on the vocabulary is

$$f_{\text{vocab}} = \frac{|V|d}{|V|d + 12Ld^2} = \frac{|V|}{|V| + 12Ld}$$

For $|V| = 32000$, $d = 512$ (so $12Ld = 6144L$):

depth L	body $12Ld^2$	+vocab $ V d$	total	f_{vocab}
4	12.58M	16.38M	28.97M	56.6%
6	18.87M	16.38M	35.26M	46.5%
8	25.17M	16.38M	41.55M	39.4%

A **pattern model** predicting in the shared latent needs no $|V|d$ output projection; with raw-byte input its embedding is $256 \cdot 512 \approx 0.13\text{M}$ (negligible). At $L = 4$ it is therefore $\approx 12.7\text{M}$ versus the token model's **28.97M** — a factor

$$28.97/12.71 = 2.28 \approx 2.3\times$$

smaller, entirely by reclaiming the vocabulary table. At small scale this reclamation is the dominant lever, not a marginal one; the " $\sim 46\%$ " headline is the $L = 6$ instance of the formula above. This is the quantitative core of the pattern-as-token efficiency claim, supported by the same-size wins of [6, 7].

5. Representational capacity

We separate two questions that are routinely conflated, differing by roughly nine orders of magnitude.

5.1 Naming capacity (combinatorial)

How many distinct patterns can a code with N nodes represent? For sparse k -of- N codes the count is $\binom{N}{k}$. Take a target of **36** billion patterns:

scheme	nodes	count	calculation
localist (1 node/concept)	3.6×10^{10}	3.6×10^{10}	one node each
sparse $k=5$ ($\approx 1.5\%$)	339	$\binom{339}{5} \approx 3.6 \times 10^{10}$	$\frac{339^5}{5!} \cdot \prod_{j=0}^4 (1 - \frac{j}{339}) \approx 3.62 \times 10^{10}$
dense binary	36	$2^{36} \approx 6.87 \times 10^{10}$	each node $\in \{0, 1\}$
16-level analog	9	$16^9 = 2^{36} \approx 6.87 \times 10^{10}$	each node $\in \{0, \dots, 15\}$

Pushing sparsity: a 2%-sparse 512-node code names

$$\binom{512}{10} = \frac{\prod_{j=0}^9 (512 - j)}{10!} \approx \frac{512^{10} e^{-45/512}}{3.6288 \times 10^6} \approx \frac{(1.24 \times 10^{27})(0.916)}{3.63 \times 10^6} \approx 3.1 \times 10^{20}$$

distinct patterns. Naming is not the bottleneck.

5.2 Johnson–Lindenstrauss: near-orthogonal packing is exponential in d

Naming assumes symbolic distinctness; for a *distributed* code we care how many patterns are mutually **distinguishable** (near-orthogonal). For independent uniform unit vectors $\mathbf{u}, \mathbf{v} \in \mathcal{S}^{d-1}$, the inner product concentrates at 0 with sub-Gaussian tails,

$$\Pr[|\langle \mathbf{u}, \mathbf{v} \rangle| > \epsilon] \leq 2 \exp\left(-\frac{d\epsilon^2}{2}\right).$$

By the probabilistic method, sampling n random vectors and union-bounding over $\binom{n}{2} < n^2/2$ pairs, all pairwise inner products are $\leq \epsilon$ with positive probability whenever $n^2 \exp(-d\epsilon^2/2) < 1$, i.e.

$$n \lesssim \exp\left(\frac{\epsilon^2 d}{4}\right).$$

The packing is exponential in d , with a base set by the tolerance ϵ (note the per-pair standard deviation is $1/\sqrt{d}$, so for $d = 512$ it is ≈ 0.044 and ϵ is measured in a few of these units):

ϵ ($\max \langle \cdot, \cdot \rangle $)	$n \approx e^{\epsilon^2 d/4}, d=512$
0.2	1.7×10^2
0.3	1.0×10^5
0.4	7.9×10^8
0.5	7.9×10^{13}

This is a (loose) existence lower bound; it establishes the *headroom* that makes "hold many concepts in superposition and separate them at readout" quantitatively plausible rather than cramped. Tighter spherical-code constructions do better.

5.3 Storage capacity: naming $\neq$ storing

How many patterns can be stored as reliable attractors, so that a noisy cue for concept c retrieves ξ_c ? This is a strictly harder question, and the classical answer is sobering. The Amit–Gutfreund–Sompolinsky analysis of the Hopfield model [9] gives a critical capacity

$$P_{\max} \approx 0.138 N.$$

To store $P = 3.6 \times 10^{10}$ attractors classically would require

$$N = \frac{P}{0.138} = \frac{3.6 \times 10^{10}}{0.138} \approx 2.6 \times 10^{11} \text{ neurons},$$

i.e. ~ 260 billion neurons and $N^2/2 \approx 3.4 \times 10^{22}$ weights — infeasible by any measure. **If "store concepts as attractors" meant classical Hopfield, the entire capacity story would collapse:** hundreds of billions of nodes to store what a few hundred can name.

5.4 Resolution: modern Hopfield = attention

Modern Hopfield networks [10] replace the quadratic energy with a log-sum-exp energy, yielding **exponential** storage capacity in the representation dimension (for well-separated binary patterns, $\sim 2^{d/2}$), retrieval in a **single** update, and an update rule identical to attention:

$$\text{retrieve}(\xi) = X \text{softmax}(\beta X^\top \xi), \quad \beta = 1/\sqrt{d},$$

with X the stored patterns as columns. Inverting the capacity bound, storing $P = 3.6 \times 10^{10}$ patterns needs only

$$2^{d/2} \geq P \iff d \geq 2 \log_2 P = 2 \times 35.07 \approx 70,$$

so a representation of a few *hundred* dimensions (for margin) suffices, with an associative matrix of d^2 weights ($\approx 0.26\text{M}$ for $d = 512$, ~ 1 MB in fp32) that is **independent of the number of stored patterns** because patterns superpose additively. The capacity lives in the superposition, not in an enumerated table; concepts are never listed. This resolution — keep the combinatorial *naming* intuition, reject the classical **0.138N storage** model, adopt modern-Hopfield/attention — is the crux of the capacity analysis and directly connects C3 to the same attention primitive used elsewhere in the model.

6. Frequency salience ("glow"): calibrated confidence

6.1 Hebbian deepening and a Born-consistent prior

On each encounter of concept c with sparse pattern ξ_c we update

$$W \leftarrow W + \eta \xi_c \xi_c^\top, \quad n_c \leftarrow n_c + 1,$$

deepening c 's basin (Hebbian well-deepening) and incrementing an evidence count. We define **brightness** by a Weber–Fechner (log) law with salience weighting and homeostatic normalization (§6.3):

$$\pi_c = \frac{\log(1 + n_c) \cdot \text{idf}(c)}{\sum_{c'} \log(1 + n_{c'}) \cdot \text{idf}(c')}.$$

Brightness is simultaneously a **prior**. To make prior probability track brightness, $p(c) \propto \pi_c$, the *amplitude* must scale as

$$|\alpha_c| \propto \sqrt{\pi_c},$$

because the Born rule gives $p(c) = |\alpha_c|^2$. The glow layer is thus Born-consistent by construction rather than bolted on.

6.2 Confidence as a retrieval margin

Brightness is a prior; **confidence** is decisiveness of retrieval — the top-1/top-2 margin,

$$\text{conf} = \pi_{(1)} - \pi_{(2)},$$

which is directly inspectable: for any emission one can print how close the runner-up was. A usable abstention rule is a single inequality: **emit if conf > τ , else hedge**. For an on-device model that cannot defer to a larger cloud model, a built-in, calibrated "I don't know" is the product.

6.3 The frequency-runaway instability and its mandatory fix

Naive brightness $\pi_c \propto n_c$ is unstable: the most frequent token dominates. In English the argmax is the space character or *the* ($n \approx 10^6$), which would become the brightest, deepest, highest-prior concept — the model would be "confident" everywhere. Three components, **all mandatory and on by default**, remove this:

1. **Log-compression** $\log(1 + n_c)$ (Weber–Fechner). A token seen 10^6 times is only $\log(10^6)/\log(10^3) = 2\times$ brighter than one seen 10^3 times, not $10^3\times$.
2. **Divisive/homeostatic normalization** $\sum_c \pi_c = 1$ (with an optional temporal decay $n_c \leftarrow \lambda n_c$), bounding total brightness so raising one concept lowers others.
3. **IDF-like salience** $\text{idf}(c) = \log(\text{contexts}/\text{contexts}(c))$: a concept firing in *every* context is uninformative and receives salience $\rightarrow 0$; a context-specific concept is salient.

Omitting any one restores the runaway. This is not tuning; it is load-bearing — the difference between a calibrated signal and a model loudly certain about a blank space.

6.4 Calibration metric

We evaluate calibration with the **Expected Calibration Error** over M confidence bins $\{B_m\}$:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|.$$

The product claim reduces to one testable inequality (H5): a glow model must achieve $\text{ECE}(\text{glow}) < \text{ECE}(\text{no-glow})$ on the **held-out** split; otherwise brightness is uncorrelated with correctness and we narrow the claim from "calibrated confidence" to "salience weighting."

7. Why classical, not a quantum computer

Three walls and one structural feature make a fault-tolerant quantum computer the *wrong* tool for this workload.

7.1 Barren plateaus

For wide parameterized quantum circuits that approximate **2**-designs, the loss-gradient variance vanishes exponentially in the qubit count n [11]:

$$\text{Var}[\partial_\theta L] \sim 2^{-\alpha n}, \quad \alpha > 0.$$

The consequence is a *sampling* catastrophe. Estimating a gradient component from measurements incurs shot noise of order $1/\sqrt{M}$ for M circuit repetitions; resolving a signal whose standard deviation is $\sim 2^{-\alpha n/2}$ to fixed signal-to-noise requires

$$M \gtrsim \frac{1}{\text{Var}[\partial_\theta L]} \sim 2^{\alpha n}$$

repetitions — exponentially many measurements exactly where a language model needs many parameters. Our classical complex network has ordinary, exact (to float precision) gradients with no such scaling.

7.2 Dequantization

A sequence of dequantization results [12] shows that several flagship QML "exponential speedups" (recommendation systems, low-rank linear algebra, PCA, certain kernels) admit classical algorithms only *polynomially* slower, once the classical algorithm is granted the same sample-and-query access the quantum algorithm implicitly assumes. For the dense, data-rich, approximately-low-rank regime of language modeling, the burden of proof for a quantum advantage is unmet.

7.3 The one proven advantage is not language

The rigorous exception [13] constructs a dataset for which, *assuming the classical hardness of discrete logarithm*, no classical learner beats random guessing by an inverse polynomial, while a quantum-kernel SVM classifies accurately and robustly. It is a genuine theorem — and it is a cryptographically structured,

data-scarce classification problem, not natural language. Language has no discrete-log structure and is not data-scarce.

7.4 The simulability pincer

The deepest obstruction is a pincer [14]:

trainable (plateau-free) $\Rightarrow$ classically simulable $\Rightarrow$ no quantum advantage,

not classically simulable $\Rightarrow$ barren plateau $\Rightarrow$ not trainable.

The regime that is trainable is the regime a classical machine can already reproduce. If the model trains at all, run it classically.

7.5 Measurement collapse: the screenshot is a classical feature

A central deliverable (§5, §6) is the **whole-state screenshot**: read every amplitude, phase, brightness, and effective-rank statistic for inspection and calibration. On real quantum hardware this is impossible — measurement collapses the state to a single sample, and full-state tomography costs a number of measurements exponential in system size. On a classical machine ψ is an array of floats, readable non-destructively at any time. **The inspectability that is the product is a property the classical implementation has and the quantum one cannot.** This is a reason to prefer classical, not a concession.

8. Complexity and hardware budget

8.1 Arithmetic overhead of complex layers

A real linear map $y = Wx$, $W \in \mathbb{R}^{d \times d}$, costs d^2 multiplies and $d(d-1)$ adds, $\approx 2d^2$ FLOPs. A complex multiply requires **4** real multiplies and **2** real adds (naive) or **3** multiplies and **5** adds (Karatsuba/Gauss). Counting multiplies — the dominant cost — a complex-valued layer at equal width d costs

$$\text{FLOP}_{\mathbb{C}} \approx (3-4) \times \text{FLOP}_{\mathbb{R}}.$$

Unitary parameterization adds an $O(d^3)$ term (Cayley) or $O(d^2)$ (Givens) per unitary layer. All overheads are polynomial, deterministic, and — as shown next — M1-affordable. Fair comparisons therefore hold *parameters* equal and report cost on the Pareto axis (§9).

8.2 The generation wall and its mitigations

Born-rule generation computes $p_k = |\langle e_k | O\psi \rangle|^2$ for every $k \in V$: an output map O of $|V|d$ parameters and $|V|d$ MACs per step. For $|V| = 32000$, $d = 512$ that is **16.38M** MACs and **16.38M** parameters **per step** — the same wall a large-vocabulary softmax faces. Mitigations (each an ablation), with concrete savings:

mitigation	mechanism	params / MACs per step	factor vs 16.38M
full Born	$O \in \mathbb{R}^{ V \times d}$	16.38M	1×
low-rank $r=64$	$O = UV^\top, r(V + d)$	2.08M	7.9×
VQ codebook $K=1024$	readout over K codes, Kd	0.52M	31×
hierarchical	balanced tree, $d \log_2 V $	7.66k	$\sim 2140\times$
tied I/O	reuse input table	0 extra	(removes 16.38M)

Coconut-style feedback [6] further **amortizes** the wall: feeding the continuous predicted pattern back in runs generation in pattern space and pays the $O(|V|d)$ readout only at emission points, not every internal step. If no mitigation reaches parity, we narrow to encoder-only tasks (classification/retrieval), where the evidence [1,3] is already strong.

8.3 Parameter and memory budget (the M1 fit)

Using §4.4 at $L = 4$, $d = 512$: the pattern model is ≈ 12.6 M parameters. Its training footprint in fp32 with Adam (which stores first and second moments m, v):

$$\underbrace{12.6\text{M} \times 4\text{B}}_{\text{params } 50.4\text{ MB}} + \underbrace{2 \times 12.6\text{M} \times 4\text{B}}_{\text{Adam } m,v \text{ } 100.8\text{ MB}} \approx 151\text{ MB},$$

plus gradients (≈ 50 MB) and batch-dependent activations. The associative memory is a single $d \times d$ matrix, $512^2 \times 4\text{B} \approx 1.0$ MB, independent of concept count. Explicit codebooks, if materialized, scale as $Kd \times 4$ B: $K = 10^4 \Rightarrow 0.02$ GB, $K = 10^6 \Rightarrow 2.0$ GB (the latter is the M1 ceiling — hence we default to the superposed matrix and small K , and never materialize 3.6×10^{10} of anything). All figures sit within an 8–16 GB unified-memory M1.

8.4 Training-time estimate (shown, not asserted)

Using the standard training-FLOP estimate $C \approx 6ND$ (N parameters, D tokens processed) and an M1 GPU throughput of $\sim 2.6 \times 10^{12}$ FP32 FLOP/s:

- **AG News** ($N \approx 2.4$ M as in [1]; $D \approx 1.2 \times 10^5$ samples $\times 50$ tok $\times 10$ ep $= 6 \times 10^7$):
 $C \approx 6(2.4 \times 10^6)(6 \times 10^7) = 8.6 \times 10^{14}$ FLOP $\Rightarrow \approx 330$ s ≈ 5.5 min; at $N \approx 12.6$ M, ≈ 29 min.
Estimate range $\sim 6\text{--}15$ min for the small models we run.
- **Char/byte LM** ($N \approx 12.6$ M; $D \approx 10^8$): $C \approx 6(12.6 \times 10^6)(10^8) = 7.6 \times 10^{15}$ FLOP $\Rightarrow \approx 48$ min;
longer schedules push to $\sim 50\text{--}125$ min.

These are engineering budgets derived from FLOP counts, to be replaced by measured wall-clock under logged configs (they are not empirical results). Their purpose is to show that sweeping ≥ 5 seeds $\times$ several ablations is a *single-day* activity on one machine — a precondition for the fairness protocol of §9.

8.5 The one real gotcha: Metal

Apple Metal has only partial complex support. We (i) treat the CPU (NdArray) backend as the source of truth for bit-for-bit-reproducible headline numbers, and (ii) represent all complex state as **(Re, Im)** pairs so that, by Proposition 1, unitary complex ops become structured *orthogonal real* ops the GPU handles natively. Metal is a stopwatch (Pareto/latency), never the witness for a headline number.

9. Experimental protocol and falsification

9.1 Five hypotheses, each with a kill number

#	hypothesis	metric	kill number
H1	complex phase carries information a real net cannot recover at equal params	accuracy gap: full-complex vs phase=0 ablation (param-matched)	gap $\leq 1\sigma$ of seed noise $\Rightarrow$ drop "quantum"
H2	next-pattern is competitive with next-token at equal params	val perplexity (fixed decoder) vs param-matched token baseline	pattern $> 10\%$ worse or collapse $\Rightarrow$ stop project
H3	pattern prediction does not collapse	effective rank / per-dim variance	erank $< 0.5d$ or variance within $10\times$ constant $\Rightarrow$ stop
H4	(headline) invariance — a concept re-triggers nearly the same pattern across <i>surface variation</i> : byte-noise, augmentation, and paraphrase	node-overlap hit-rate, clean cue vs varied cue; overlap monotone in semantic similarity	hit-rate $< \sim 90\%$ or overlap not monotone $\Rightarrow$ narrow to soft retrieval
H5	brightness tracks correctness (calibration)	ECE, glow vs no-glow on held-out	ECE(glow) $\geq$ ECE(no-glow) $\Rightarrow$ narrow to salience

H4 is the **headline** claim (the invariance/convergence property the encoder is named for); H1 governs whether the *wave/phase* mechanism earns its parameters; H2/H3 govern whether the pattern objective is a viable *project* at all; H5 governs the confidence product. Each fails to a specific, pre-registered number. The stronger, later form of H4 — invariance across *modalities* (the same concept from an image, a sentence, or a video landing on nearly the same pattern) — requires from-scratch multi-modal encoders and is a **scaling destination** (§10), deliberately excluded from the decision-grade claim because it is not demonstrable under C1 on an M1.

9.2 Fairness controls

(1) **Equal parameters** ($\pm 5\%$), verified by an automated counter before any run. (2) **Equal tuning budget** — identical hyperparameter-search size and early-stopping rule for both arms. (3) **Equal data, splits, loop** — fixed, hashed, leak-checked splits; same optimizer, schedule, and compute. (4) **One switch per comparison**

— named ablations `phase=0`, `no_unitary`, `no_interference`, `no_modulation`, `no_glow`, `bpe` (subword is an ablation, never a default). (5) **Full config with every number** — seed, versions, git SHA; reproduces bit-for-bit on CPU. (6) **Seeds as a distribution** — report mean $\pm\sigma$ over ≥ 5 seeds.

9.3 The statistical decision

For H1 the estimand is $\Delta = \text{metric}(\text{complex}) - \text{metric}(\text{phase} = 0)$. Let s be the pooled per-seed standard deviation. We pre-register a one-sided decision at effect size $\Delta > s$ (a "win" smaller than one seed- σ is treated as noise). A supporting diagnostic — the phase histogram — must show phase is *used*; if the free-phase model drives $\phi \rightarrow 0$ on its own, that is itself an H1 kill (the data did not want phase). The same σ -thresholded discipline applies to every headline comparison; the threshold is written down before the run to prevent goalpost-moving.

9.4 What "test against LLMs" means

It does **not** mean a head-to-head against frontier models; that comparison is uninformative across a $\sim 10^3 \times$ scale gap and any framing implying we might win it is a credibility hazard. It means three apples-to-apples things: **(a) mechanism parity** — compare against a *parameter-matched Transformer we build ourselves* at equal data and budget; **(b) the Pareto frontier** — plot quality against cost (parameters, FLOPs, memory, latency, energy) across several small sizes; **(c) scaling extrapolation** — fit the small-scale trend and extrapolate with intervals, explicitly labeled as extrapolation, never presented as a measured large-scale result.

10. Limitations and threats to validity

1. **Generation at scale is unproven.** Every quantum-inspired win we cite [1,3] is classification/matching; the $O(|V|d)$ Born-readout wall is unbroken at large vocabulary. §8.2 offers mitigations and an honest fallback (encoder-only), not a solution.
2. **The phase advantage may not survive a fair ablation.** [1]'s reference used pretrained embeddings; a parameter- and tuning-matched real baseline may tie (H1). We have pre-committed to dropping the "quantum" framing if so.
3. **Unitary maps are linear.** A pure unitary layer cannot gate/forget; practical models reintroduce nonlinearity, partially returning the complexity the formalism promised [4]. The needed amount is an open empirical question.
4. **Anti-collapse regularizers have their own failure modes** (VICReg hyperparameter sensitivity; contrastive negative-sampling; VQ codebook collapse). H3 instruments for all of them, but "no collapse" is a continual measurement, not a one-time guarantee.
5. **Capacity bounds are existence/aggregate results.** The JL packing (§5.2) is a loose lower bound; modern-Hopfield capacity [10] depends on pattern separation Δ , and real learned patterns are neither uniform nor maximally separated.

6. **Calibration can be gamed by evaluation leakage.** ECE on training-distribution data is trivially flattering; H5 is defined on held-out splits precisely to prevent this.
 7. **Small-scale conclusions may not scale.** Nothing here establishes that complex/pattern/unitary structure keeps winning into the billions of parameters; §9.4 confines our claims to the frontier we can measure plus labeled extrapolation.
 8. **We enter a mature, crowded field, so the novelty is narrow by construction.** VSA/HDC is 35 years old [25,26,27,29] and shared-embedding alignment [7,28] is strong; we do not out-scale either. The contribution is precisely the *intersection* nobody occupies: a **learned, from-scratch** (C1) VSA-style encoder whose binding is realized in the wave/frequency domain, whose patterns are **learned attractors**, and which carries a **calibrated, inspectable** confidence signal. If a reviewer reads any one of these axes in isolation, the work looks derivative; the claim lives only in their conjunction, and each axis is individually ablatable (§9.2).
 9. **Cross-modal invariance is asserted, not demonstrated.** The "one pattern for image, text, or video" ambition requires from-scratch per-modality encoders that C1 and the M1 budget do not permit to validate here. Presenting it as a result — rather than as the stated destination beyond single-modality invariance (H4) — would be the exact over-claim this framing exists to avoid.
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11. Conclusion

The object of this paper is a **learned, from-scratch encoder that maps inputs to stable, surface-invariant, inspectable concept patterns** — the representational problem underneath grounding, retrieval, and on-device anti-hallucination. Its substrate is a named, 35-year-old field, Vector Symbolic Architectures, whose three primitives (bind, superpose, permute) match the intuition of a combinatorial pattern graph exactly; its one mechanistic novelty is that binding is performed in the **wave / frequency domain**, where — by Proposition 0 — circular-convolution binding is complex multiplication (phase addition), making the complex-amplitude mathematics a learnable realization of VSA binding rather than a separate thesis. Around that substrate we place an invariance objective with instrumented anti-collapse (§4), a capacity account that distinguishes *naming* from *storage* and resolves a nine-order-of-magnitude fallacy (§5), and a Bayesian–Hebbian calibrated confidence layer (§6). The efficiency and feasibility arguments are concrete — a **2.3×** parameter reduction from predicting in a shared pattern space (§4.4), an associative store whose ~ 1 MB is independent of concept count (§5.4), and a regime that decides every hypothesis on an M1 in a day (§8). The boundaries are equally concrete and stated as such: the *wave/phase* mechanism is contingent on one pre-registered ablation (H1); generation at scale is gated behind the readout wall (§8.2); and **cross-modal** invariance is a destination, not a result — the testable claim is single-modality invariance (H4). We argue separately that running classically is not a compromise but the correct engineering choice, since a fault-tolerant quantum computer is obstructed by barren plateaus, undercut by dequantization, squeezed by the simulability pincer, and structurally unable to deliver the inspectable full-state readout that is the product (§7). The contribution is not a benchmark victory but a *decidable* research program — five hypotheses, five kill numbers, one machine — that can be run to a definite yes or no.

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Appendix A — Notation

symbol	meaning
$\psi \in \mathbb{C}^d$	complex state; stored as (Re ψ , Im ψ)
$\alpha = r e^{i\phi}$	single amplitude: magnitude r , phase ϕ
$\langle a b \rangle$	Hermitian inner product $\sum_k \overline{a_k} b_k$
$\odot$	Hadamard (element-wise) product (= wave-domain bind)
$\otimes$	circular convolution (VSA bind ; = $\odot$ in the frequency domain, §3.3)
$\mathcal{F}, P$	discrete Fourier transform; a fixed permutation (VSA permute)
U, H, A	unitary map; Hermitian generator; (skew-Hermitian) Cayley argument
$z, \hat{z}$	target pattern; predicted pattern
$\text{sg}(\cdot)$	stop-gradient
$\text{erank}(Z)$	effective rank (participation ratio) of Z
N, k	code size; number of active (sparse) nodes
π_c, n_c	brightness of concept c ; its encounter count
$ V , d, L$	vocabulary size; model width; depth

Appendix B — Key numerical results at a glance

quantity	value	source
Wave Network single-layer AG News (interference / modulation)	90.91% / 91.66%	[1], §2
gain over single Transformer layer	+19.23 / +19.98 pts	[1]
vocabulary parameter fraction, $L=6$	46.5%	§4.4
pattern vs token model size, $L=4$	2.3× smaller	§4.4
2% -sparse 512 -node naming capacity	$\binom{512}{10} \approx 3.1 \times 10^{20}$	§5.1
near-orthogonal packing, $d=512$, $\epsilon=0.5$	$\sim 7.9 \times 10^{13}$	§5.2
classical Hopfield nodes to store 3.6×10^{10}	$\sim 2.6 \times 10^{11}$	§5.3
modern-Hopfield dimension to store 3.6×10^{10}	$d \gtrsim 70$	§5.4
associative matrix size, $d=512$	~ 1 MB	§8.3
pattern-model training footprint	~ 151 MB	§8.3
barren-plateau measurement cost	$M \sim 2^{an}$	§7.1
complex-layer arithmetic overhead	3–4× real	§8.1
Born-generation readout cost	$O(V d) = 16.38\text{M MAC/step}$	§8.2